

1 The multiplicity boundary floor

This section proves an exact lower bound on the relative control defect of the theta-generated representatives. The bound counts every retained zero with its complete multiplicity. The source constructions are A1644, lines 2576–2718, and A1771, lines 4477–4835, in the authoritative `audit_segment.U0041.U0054.md`. The rank-two identity itself is A1771, lines 4688–4725. The invariant trace and the invariant-subspace argument below are new deductions from those source constructions.

All matrix coordinates on the packet algebra are the original polynomial coordinates. The generator is $D = -x\partial_x$, the arithmetic multiplier is $g = 2\xi$, and the Hilbert-space measure is dx on the positive half-line. The local extension unit, the unscaled orthogonal vectors, their norm squares, and all nilpotent jets are retained.

1.1 The original source, packet algebra, and arithmetic section

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(s) = 2\xi(s),$$

with the entire continuations understood at the apparent singularities. Let

$$\begin{aligned} V &= \left\{ \phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0 \right\}, \\ \mathcal{B} &= \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \\ D &= -x\partial_x, \quad (\Theta\phi)(x) = \sum_{a \neq 0} \phi(ax) = 2 \sum_{a \geq 1} \phi(ax). \end{aligned} \tag{BF.01}$$

Here $\mathcal{S}(\mathbb{R})$ is the Schwartz space, and the summation indices a in Θ are integers. The symbol $\mathcal{B}$ denotes exactly the strong-Schwartz target specified in A1694, lines 3049–3073. Put

$$Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H}_+ = L^2(\mathbb{R}_{>0}, dx).$$

The scalar product convention is

$$\langle F, H \rangle_+ = \int_0^\infty \overline{F(x)} H(x) dx,$$

antilinear in the first argument. Adjoint signs on maps into $\mathcal{H}_+$ use this scalar product and the standard scalar product in the stated polynomial coordinates on their finite-dimensional domains.

Retain the actual source function and its derivative sequence:

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \\ f_0 &= \Theta\phi_*, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta\phi_j = D^j f_0 \quad (j \geq 0), \\ (\mathcal{M}F)(s) &= \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \mathcal{M}f_0 = g. \end{aligned} \tag{BF.02}$$

The source identity $\mathcal{M}(D^j f_0)(s) = s^j g(s)$, with these exact constants, is A1644, lines 2566–2574, and A1771, lines 4479–4529. The commutation $D\Theta\phi = \Theta D\phi$ uses the same generator on the source and target. Differentiation of the defining series is justified on each compact subinterval of $(0, \infty)$ by the Schwartz estimates.

Let Z be a nonempty finite set of actual zeros of g . For $\rho \in Z$, let $m_\rho \geq 1$ be its complete order of vanishing. Define

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z = \sum_{\rho \in Z} m_\rho, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t : E_Z \longrightarrow E_Z. \quad (\text{BF.03})$$

The ordered coordinates on E_Z are the classes of $1, t, \dots, t^{d-1}$. Multiplication by t is not replaced by a shifted generator. The empty packet is treated separately at the end.

For a function holomorphic near every member of Z , j_{h_Z} means its complete collection of jets of orders $0, \dots, m_\rho - 1$, identified with E_Z by the Chinese remainder isomorphism. In particular, the finite Mellin-jet map is

$$J_Z : \mathcal{B} \longrightarrow E_Z, \quad J_Z F = j_{h_Z}(\mathcal{M}F).$$

In the local coordinate $z = s - \rho$, the ρ -component is

$$\sum_{j=0}^{m_\rho-1} \frac{(\mathcal{M}F)^{(j)}(\rho)}{j!} z^j \pmod{z^{m_\rho}}.$$

The map retains the factorials and the full jet length.

The actual extension unit is

$$\varepsilon_Z = (j_{h_Z}(g/h_Z))^{-1} = j_{h_Z}(h_Z/g) \in E_Z^\times. \quad (\text{BF.04})$$

To see why it is a unit, write

$$g(\rho + z) = z^{m_\rho} w_\rho(z), \quad w_\rho(0) \neq 0.$$

The polynomial $h_Z(\rho + z)$ has precisely the same zero order. Thus g/h_Z and h_Z/g are mutually inverse holomorphic units near every selected zero. Equation (BF.04) keeps their complete jets. This finite-algebra unit is distinct from the supported-zero scalar e of the original split-support construction.

For $u \in E_Z$, let $R_Z(u)$ be the unique polynomial of degree less than d representing $\varepsilon_Z u$, and define the extension row

$$\ell_Z : E_Z \longrightarrow \mathbb{C}, \quad \ell_Z(u) = [t^{d-1}]R_Z(u). \quad (\text{BF.05})$$

The established arithmetic section in A1644, lines 2594–2718, and A1771, lines 4499–4510, is the map

$$s_Z : E_Z \longrightarrow \mathcal{B}$$

with the identities

$$J_Z s_Z = I_{E_Z}, \quad J_Z \Theta = 0, \quad q s_Z = \sigma_Z : E_Z \hookrightarrow \mathcal{Q}, \quad D s_Z - s_Z A = f_0 \ell_Z. \quad (\text{BF.06})$$

These are the inherited source identities used below, with the actual source section and its analytic construction. For a one-point packet at a simple zero, their coefficient is exactly

$$\ell_Z(1) = \frac{1}{g'(\rho)} = \frac{1}{2\xi'(\rho)}.$$

No replacement of this coefficient by a unit-free scalar is made.

1.2 Theta relation spaces and the exact rank-two boundary

Define

$$V_n = \text{span}_{\mathbb{C}}\{\phi_0, \dots, \phi_n\}, \quad L_n = \Theta V_n = \text{span}_{\mathbb{C}}\{f_0, \dots, f_n\}, \quad n \geq 0. \quad (\text{BF.07})$$

The vectors f_j are linearly independent. Indeed, a finite dependence has Mellin transform $g(s)P(s) = 0$, where $P(s) = \sum_j c_j s^j$. Since g is not identically zero, it is nonzero on a nonempty open set. Therefore P vanishes on that open set and is the zero polynomial. Every coefficient c_j is zero.

Let $F_n : \mathbb{C}^{n+1} \rightarrow \mathcal{H}_+$ have columns $f_0, \dots, f_n$, and set

$$H_n = F_n^* F_n, \quad P_n = F_n H_n^{-1} F_n^*, \quad R_n = (I - P_n) s_Z, \quad G_n = R_n^* R_n. \quad (\text{BF.08})$$

Independence gives $H_n \succ 0$. Direct multiplication gives $P_n^2 = P_n = P_n^*$, and its range is L_n ; it is the actual orthogonal projection onto the specified finite theta relation space. The expression

$$b_n(u) = \sum_{j=0}^n (H_n^{-1} F_n^* s_Z u)_j \phi_j$$

defines a map $b_n : E_Z \rightarrow V_n$ satisfying $P_n s_Z = \Theta b_n$. Consequently, by (BF.06),

$$qR_n = \sigma_Z, \quad J_Z R_n = I_{E_Z}, \quad G_n = s_Z^* s_Z - s_Z^* F_n H_n^{-1} F_n^* s_Z \succ 0. \quad (\text{BF.09})$$

For the strict inequality, $v^* G_n v = \|R_n v\|_+^2 = 0$ implies $R_n v = 0$ and hence $v = J_Z R_n v = 0$. Thus all inverses of G_n used below are inverses of strictly positive matrices at the stated finite level.

The original theta boundary and the associated control matrix are

$$\begin{aligned} B_n &= D R_n - R_n A = \Theta K_n, \\ K_n &= \phi_* \ell_Z - D_V b_n + b_n A, \\ W_n &= A^* G_n + G_n A - G_n. \end{aligned} \quad (\text{BF.10})$$

Here D_V is D on V . The boundary formula follows by substituting $R_n = s_Z - \Theta b_n$ into (BF.06). It uses only original theta boundaries.

For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle_+ + \langle F, DH \rangle_+ = - \left[x \overline{F(x)} H(x) \right]_0^\infty + \langle F, H \rangle_+ = \langle F, H \rangle_+. \quad (\text{BF.11})$$

The defining estimates of $\mathcal{B}$ make the boundary expression zero at both endpoints. Applying (BF.11) to columns of R_n , and substituting $DR_n = R_n A + B_n$, proves

$$W_n = -(R_n^* B_n + B_n^* R_n). \quad (\text{BF.12})$$

The coefficient of G_n in (BF.10) is exactly one because the measure in (BF.11) is dx . Even extension to the full line has squared norm twice this norm; it is not silently substituted here.

Construct the unscaled orthogonal vectors

$$u_0 = f_0, \quad u_n = (I - P_{n-1})f_n \quad (n \geq 1), \quad \mathfrak{h}_n = \langle u_n, u_n \rangle_+ > 0, \quad r_n = \mathfrak{h}_n^{-1} u_n^* s_Z : E_Z \rightarrow \mathbb{C}. \quad (\text{BF.13})$$

The operator $u_n^* : \mathcal{H}_+ \rightarrow \mathbb{C}$ sends F to $\langle u_n, F \rangle_+$. Each u_n has leading coefficient one on f_n ; none is divided by its norm. Since the f_j are real functions, their Gram matrices are real, and each u_n is real.

Degree and orthogonality determine the recurrence

$$\begin{aligned} Du_0 &= u_1 + \frac{1}{2}u_0, \\ Du_n &= u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1). \end{aligned} \quad (\text{BF.14})$$

Here is the complete coefficient calculation. As $u_n = f_n + \sum_{j < n} c_j f_j$, Du_n belongs to L_{n+1} and has coefficient one on u_{n+1} . For $j \leq n-2$, (BF.11) gives

$$\langle u_j, Du_n \rangle_+ = \langle u_j, u_n \rangle_+ - \langle Du_j, u_n \rangle_+ = 0,$$

because $Du_j \in L_{j+1} \subseteq L_{n-1}$. Reality and (BF.11) give

$$2\langle u_n, Du_n \rangle_+ = \mathfrak{h}_n,$$

so the full diagonal coefficient is $1/2$, not merely its real part. For $n \geq 1$,

$$\langle u_{n-1}, Du_n \rangle_+ = -\langle Du_{n-1}, u_n \rangle_+ = -\mathfrak{h}_n.$$

Dividing only this coefficient by the corresponding original norm square $\mathfrak{h}_{n-1}$ yields the last term of (BF.14).

The section coefficients satisfy

$$P_n s_Z = \sum_{j=0}^n u_j r_j, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} r_{n+1} = r_n \left(\frac{1}{2}I - A \right) + r_{n-1} - \mathbf{1}_{\{n=0\}} \ell_Z, \quad r_{-1} = 0. \quad (\text{BF.15})$$

The indicator $\mathbf{1}_{\{n=0\}}$ equals one for $n = 0$ and zero otherwise. To check the row equation, pair $Ds_Z = s_Z A + u_0 \ell_Z$ against u_n . The right side is

$$\mathfrak{h}_n r_n A + \mathbf{1}_{\{n=0\}} \mathfrak{h}_0 \ell_Z.$$

By (BF.11) and (BF.14), the left side is

$$\frac{1}{2} \mathfrak{h}_n r_n - \mathfrak{h}_{n+1} r_{n+1} + \mathfrak{h}_n r_{n-1},$$

where the last term is zero for $n = 0$. Rearrangement gives (BF.15). The initial extension functional is therefore fixed by the original arithmetic section.

We now obtain the source's rank-two formula with its exact sign. Since

$$B_n = f_0 \ell_Z - DP_n s_Z + P_n s_Z A,$$

the first and third terms have image in L_n , orthogonal to the image of R_n . Equation (BF.12) becomes

$$W_n = R_n^* DP_n s_Z + (DP_n s_Z)^* R_n.$$

Equations (BF.14) and (BF.15) give

$$(I - P_n)DP_n s_Z = u_{n+1} r_n, \quad u_{n+1}^* R_n = u_{n+1}^* s_Z = \mathfrak{h}_{n+1} r_{n+1}.$$

It follows that

$$\boxed{W_n = \mathfrak{h}_{n+1} (r_{n+1}^* r_n + r_n^* r_{n+1})}. \quad (\text{BF.16})$$

Also $R_{n+1} = R_n - u_{n+1} r_{n+1}$, with R_{n+1} orthogonal to u_{n+1} . Taking Gram matrices proves the exact decrement

$$G_n - G_{n+1} = \mathfrak{h}_{n+1} r_{n+1}^* r_{n+1} \succeq 0. \quad (\text{BF.17})$$

These formulas are A1771, lines 4688–4734. The rank bound in (BF.16) concerns the boundary control form on E_Z ; the dimension d , and the full operator A , have not been reduced to two dimensions.

For clarity, the retained source-support transition is also exact. Write $Q_{V_n} = \mathcal{B}/\Theta V_n = \mathcal{B}/L_n$. Since $D\Theta V_n \subseteq \Theta V_{n+1}$, differentiation gives a well-defined map

$$\overline{D}_n : Q_{V_n} \longrightarrow Q_{V_{n+1}}, \quad [F] \longmapsto [DF].$$

Before that transition, the boundary class is

$$[B_n v]_{Q_{V_n}} = -[u_{n+1}]_{Q_{V_n}} r_n(v).$$

The layer map

$$\Theta V_{n+1}/\Theta V_n \longrightarrow L_{n+1} \cap L_n^\perp, \quad [z] \longmapsto (I - P_n)z$$

is well-defined and injective because its kernel in L_{n+1} is L_n . It is surjective because every vector in the displayed orthogonal complement is fixed by $I - P_n$. The subsequent support transport makes this particular class the supported zero in $Q_{V_{n+1}}$. Its earlier pairing in (BF.16) is retained. These are the actual maps of A1771, lines 4736–4773.

1.3 Relative control spectrum and its invariant trace

For $G_n \succ 0$, define its least two-sided control excess by

$$\epsilon_n = \inf \{ \epsilon \geq 0 : -\epsilon G_n \preceq W_n \preceq \epsilon G_n \}. \quad (\text{BF.18})$$

Here $X \preceq Y$ means that $Y - X$ is a positive semidefinite Hermitian matrix. The operator $G_n^{-1}W_n$ is self-adjoint in the positive inner product $\langle u, v \rangle_{G_n} = u^* G_n v$:

$$\langle u, G_n^{-1}W_n v \rangle_{G_n} = u^* W_n v = \langle G_n^{-1}W_n u, v \rangle_{G_n}.$$

Thus ϵ_n is the largest absolute value of its eigenvalues. This uses the actual metric and operator and makes no assumption that A is normal for this metric.

In the unchanged coordinates, set

$$\mathbf{a}_n = r_n G_n^{-1} r_n^*, \quad \mathbf{b}_n = r_{n+1} G_n^{-1} r_{n+1}^*, \quad \mathbf{c}_n = r_n G_n^{-1} r_{n+1}^*. \quad (\text{BF.19})$$

The exact maps

$$\begin{aligned} \mathcal{T}_n : E_Z &\longrightarrow \mathbb{C}^2, & \mathcal{T}_n v &= (r_n v, r_{n+1} v), \\ \mathcal{U}_n : \mathbb{C}^2 &\longrightarrow E_Z, & & \\ \mathcal{U}_n(a, b) &= \mathfrak{h}_{n+1} (G_n^{-1} r_{n+1}^* a + G_n^{-1} r_n^* b) \end{aligned} \quad (\text{BF.20})$$

satisfy

$$\mathcal{U}_n \mathcal{T}_n = G_n^{-1} W_n, \quad \mathcal{T}_n \mathcal{U}_n = \mathfrak{h}_{n+1} \begin{pmatrix} \mathbf{c}_n & \mathbf{a}_n \\ \mathbf{b}_n & \mathbf{c}_n \end{pmatrix}.$$

For every nonzero eigenvalue, applying $\mathcal{T}_n$ and $\mathcal{U}_n$ gives mutually inverse eigenspace maps up to that nonzero scalar. Therefore the nonzero eigenvalues of the two composites agree. Cauchy–Schwarz for the positive form on row covectors induced by G_n^{-1} gives $\mathbf{a}_n \mathbf{b}_n \geq |\mathbf{c}_n|^2$. The two roots of the displayed 2-by-2 characteristic polynomial are consequently

$$\lambda_{n,\pm} = \mathfrak{h}_{n+1} \left(\text{Re } \mathbf{c}_n \pm \sqrt{\mathbf{a}_n \mathbf{b}_n - (\text{Im } \mathbf{c}_n)^2} \right), \quad \lambda_{n,+} \geq 0 \geq \lambda_{n,-}. \quad (\text{BF.21})$$

Zero roots are included in this notation even when the corresponding eigenspace does not occur in E_Z . There are no other nonzero eigenvalues by the factorization (BF.20). Thus

$$\epsilon_n = \mathfrak{h}_{n+1} \left(|\text{Re } \mathbf{c}_n| + \sqrt{\mathbf{a}_n \mathbf{b}_n - (\text{Im } \mathbf{c}_n)^2} \right). \quad (\text{BF.22})$$

This is the exact expression in A1771, lines 4775–4821. In particular, G_n^{-1} remains present in every relative coefficient.

The trace of this control operator is independent of the chosen positive representative metric. Cyclicity of finite-dimensional trace and (BF.10) give

$$\begin{aligned} \text{tr}(G_n^{-1}W_n) &= \text{tr}(G_n^{-1}A^*G_n) + \text{tr } A - d \\ &= 2 \text{Re } \text{tr } A - d \\ &= \sum_{\rho \in Z} m_\rho (2 \text{Re } \rho - 1). \end{aligned} \quad (\text{BF.23})$$

For the final equality, the Chinese remainder decomposition preserves the local coordinates $z = t - \rho$. On the length- m_ρ summand, multiplication by t is $\rho I + M_z$. In the ordered basis $1, z, \dots, z^{m_\rho-1}$,

M_z has zero diagonal and trace zero. Thus its full trace contribution is $m_\rho \rho$. No generalized eigenvector has been deleted.

Taking the trace directly in (BF.16), and using (BF.19), yields the exact row constraint

$$\boxed{2\mathfrak{h}_{n+1} \operatorname{Re} c_n = \sum_{\rho \in Z} m_\rho (2 \operatorname{Re} \rho - 1).} \quad (\text{BF.24})$$

Define the reflection $\iota \rho = 1 - \bar{\rho}$. A reflection-stable packet means that $\iota Z = Z$ and that $m_{\iota \rho} = m_\rho$ at every member. For such a packet the contributions of every two-point orbit in (BF.23) cancel. At a fixed point of ι , $\operatorname{Re} \rho = 1/2$, so its contribution is zero. Since $\mathfrak{h}_{n+1} > 0$,

$$\begin{aligned} \operatorname{tr}(G_n^{-1} W_n) &= 0, & \operatorname{Re} c_n &= 0, \\ \lambda_{n,+} &= \epsilon_n = -\lambda_{n,-}, \\ \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathfrak{a}_n \mathfrak{b}_n - (\operatorname{Im} c_n)^2}. \end{aligned} \quad (\text{BF.25})$$

If $W_n \neq 0$, the two nonzero eigenvalues are ϵ_n and $-\epsilon_n$, each once. If $\epsilon_n = 0$, the self-adjoint operator $G_n^{-1} W_n$ has only the zero eigenvalue and hence is zero. Trace cancellation alone does not set the matrix W_n to zero.

1.4 The invariant-subspace projection argument

The next estimates hold for an arbitrary actual packet Z ; reflection stability will be used only for their final combination. Define

$$Z_+ = \{\rho \in Z : \operatorname{Re} \rho > 1/2\}, \quad Z_- = \{\rho \in Z : \operatorname{Re} \rho < 1/2\}.$$

Let

$$E_+ = \bigoplus_{\rho \in Z_+} \mathbb{C}[z]/(z^{m_\rho}), \quad E_- = \bigoplus_{\rho \in Z_-} \mathbb{C}[z]/(z^{m_\rho}). \quad (\text{BF.26})$$

The inclusions $I_\pm : E_\pm \rightarrow E_Z$ are the actual Chinese remainder inclusions, not coordinate projections chosen to make A normal. Explicitly, put $h_\rho = (t - \rho)^{m_\rho}$ and $H_\rho = h_Z/h_\rho$. Since H_ρ and h_ρ are coprime, choose $b_\rho(t)$ with $H_\rho b_\rho \equiv 1 \pmod{h_\rho}$, and let e_ρ be the class of $H_\rho b_\rho$ in E_Z . It has residue one at the ρ -summand and zero at all other summands. The maps in (BF.26) are

$$I_\pm((u_\rho)_\rho) = \sum_{\rho \in Z_\pm} e_\rho u_\rho(t - \rho).$$

Changing the representative b_ρ changes nothing in E_Z . All original polynomial coordinates and every full local length remain in these maps.

Let $A_\pm$ be multiplication by $\rho + z$ on each local summand of $E_\pm$. The intertwining identities are

$$A I_\pm = I_\pm A_\pm. \quad (\text{BF.27})$$

In particular the images are invariant generalized eigenspaces, with their nilpotent parts unchanged.

Fix $n \geq 0$. For a nonzero $E_\pm$, define

$$G_\pm = I_\pm^* G_n I_\pm, \quad P_\pm = I_\pm G_\pm^{-1} I_\pm^* G_n. \quad (\text{BF.28})$$

The notation $G_\pm, P_\pm$ in this subsection refers to this fixed value of n . Since $I_\pm$ is injective and $G_n \succ 0$, $G_\pm \succ 0$. Multiplication gives $P_\pm^2 = P_\pm$, $P_\pm I_\pm = I_\pm$, and

$$P_\pm^* G_n = G_n P_\pm.$$

Thus $P_\pm$ is the G_n -orthogonal projection onto $\text{im } I_\pm$. When $E_\pm = \{0\}$, define $P_\pm = 0$ directly; no inverse of a nonpositive-dimensional Gram matrix is needed.

For nonzero $E_\pm$, the restricted defect is exact. By (BF.27),

$$I_\pm^* W_n I_\pm = A_\pm^* G_\pm + G_\pm A_\pm - G_\pm. \quad (\text{BF.29})$$

For nonzero $E_\pm$, cyclicity of the trace of the rectangular composites, followed by (BF.29), gives

$$\begin{aligned} \text{tr}(P_\pm G_n^{-1} W_n) &= \text{tr}(G_\pm^{-1} I_\pm^* W_n I_\pm) \\ &= 2 \text{Re tr } A_\pm - \dim E_\pm \\ &= 2 \sum_{\rho \in Z_\pm} m_\rho (\text{Re } \rho - 1/2). \end{aligned} \quad (\text{BF.30})$$

For $E_\pm = \{0\}$, the first and last expressions in (BF.30) are both zero and remain equal; the inverse-containing expression is not invoked. The last trace evaluation again uses $\rho I + M_z$ on each full local summand.

The rank-two factorization gives a stronger estimate for these compressed traces than a generic rank bound would give. Indeed, the Hermitian form $v^* W_n v$ is the pullback under $v \mapsto (r_n v, r_{n+1} v)$ of

$$\mathfrak{h}_{n+1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{BF.31})$$

This form has one positive and one negative direction. To verify the inherited inertia bound directly, a positive definite subspace for W_n must map injectively under $v \mapsto (r_n v, r_{n+1} v)$; otherwise its kernel would contain a nonzero vector on which the form is zero. Its image is a positive definite subspace of (BF.31), so has dimension at most one. The same reasoning applies to a negative definite subspace. Therefore $G_n^{-1} W_n$, self-adjoint in the G_n metric, has at most one strictly positive and at most one strictly negative eigenvalue.

Use $\lambda_{n,+}$ and $\lambda_{n,-}$ from (BF.21). Let Π_+ and Π_- be their G_n -orthogonal spectral projections when the corresponding eigenvalue is strictly positive or strictly negative. When a sign is absent, set both its eigenvalue and its projection equal to zero. The finite-dimensional spectral theorem gives

$$G_n^{-1} W_n = \lambda_{n,+} \Pi_+ + \lambda_{n,-} \Pi_-, \quad \text{rank } \Pi_\pm \leq 1. \quad (\text{BF.32})$$

All remaining spectral directions have eigenvalue zero.

Let P be any G_n -orthogonal projection. For either of the nonzero projections $\Pi_\pm$, $\Pi_\pm P \Pi_\pm$ acts on a one-dimensional subspace as a positive contraction. Positivity follows because

$$\langle x, \Pi_\pm P \Pi_\pm x \rangle_{G_n} = \langle P \Pi_\pm x, P \Pi_\pm x \rangle_{G_n} \geq 0,$$

and the upper bound follows from $\|P\Pi_{\pm}x\|_{G_n} \leq \|\Pi_{\pm}x\|_{G_n}$. Its eigenvalue belongs to $[0, 1]$. Cyclicity and $\Pi_{\pm}^2 = \Pi_{\pm}$ then prove

$$0 \leq \text{tr}(P\Pi_{\pm}) = \text{tr}(\Pi_{\pm}P\Pi_{\pm}) \leq 1. \quad (\text{BF.33})$$

The equality and both inequalities also hold when $\Pi_{\pm} = 0$. Combining (BF.32) and (BF.33) gives

$$\lambda_{n,-} \leq \text{tr}(PG_n^{-1}W_n) \leq \lambda_{n,+}. \quad (\text{BF.34})$$

For example the upper bound is $\lambda_{n,+} \text{tr}(P\Pi_+) + \lambda_{n,-} \text{tr}(P\Pi_-) \leq \lambda_{n,+}$; the lower bound follows with the two signs interchanged.

Apply (BF.34) to the actual projections P_+ and P_- in (BF.30). For every packet and every $n \geq 0$, this proves

$$\boxed{\begin{aligned} \lambda_{n,+} &\geq 2 \sum_{\rho \in Z_+} m_{\rho}(\text{Re } \rho - 1/2), \\ -\lambda_{n,-} &\geq 2 \sum_{\rho \in Z_-} m_{\rho}(1/2 - \text{Re } \rho). \end{aligned}} \quad (\text{BF.35})$$

These estimates do not require reflection stability or semisimplicity.

For a reflection-stable packet, reflection bijects Z_+ and Z_- , preserves multiplicity, and negates $\text{Re } \rho - 1/2$. Thus the two positive distance sums in (BF.35) are equal. Equation (BF.25) identifies the two control eigenvalues with ϵ_n and $-\epsilon_n$. Their combination is the claimed multiplicity boundary floor:

$$\boxed{\begin{aligned} \epsilon_n &= \mathfrak{h}_{n+1} \sqrt{\mathbf{a}_n \mathbf{b}_n - (\text{Im } \mathbf{c}_n)^2} \\ &\geq \sum_{\rho \in Z} m_{\rho} \left| \text{Re } \rho - \frac{1}{2} \right| \quad (n \geq 0). \end{aligned}} \quad (\text{BF.36})$$

The restriction traces in (BF.30) show explicitly why complete multiplicities occur. The proof includes every generalized eigenspace and uses its induced metric. No normality of A , deletion of nilpotent jets, or positivity of the arithmetic Weil form has entered.

1.5 Exact consequences for a putative off-line zero

An individual eigenvector gives a useful comparison. If $0 \neq v \in E_Z$ satisfies $Av = \rho v$, then directly from (BF.10),

$$v^* W_n v = (2 \text{Re } \rho - 1) v^* G_n v. \quad (\text{BF.37})$$

Such a vector exists in every local summand: the class $z^{m_{\rho}-1}$, embedded by e_{ρ} , is nonzero and is annihilated by $A - \rho I$. By (BF.18), (BF.37) gives $\epsilon_n \geq |2 \text{Re } \rho - 1|$. Equation (BF.36) includes the additional constraints from all other roots and their multiplicities.

For a reflected off-line pair $\{\rho, 1 - \bar{\rho}\}$, with complete multiplicity m at both points, put $\delta = \text{Re } \rho - 1/2 \neq 0$. The two points are distinct, and their contributions to the sum in (BF.36) give

$$\epsilon_n \geq 2m|\delta| = m|2 \text{Re } \rho - 1| \quad (n \geq 0). \quad (\text{BF.38})$$

The real structure and functional equation of the retained xi-function are $g(\bar{s}) = \overline{g(s)}$ and $g(1-s) = g(s)$. Composition of a local Taylor expansion with conjugation or the affine map $s \mapsto 1-s$ preserves its first nonzero degree, so these symmetries preserve the complete vanishing multiplicities. Consequently, when the four points $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$ are distinct, they form a reflection-stable packet with the same complete multiplicity m . Their exact bound is

$$\epsilon_n \geq 4m|\delta| = 2m|2\operatorname{Re} \rho - 1| \quad (n \geq 0). \quad (\text{BF.39})$$

For a collapsed symmetry orbit, only the actual distinct points with their actual multiplicities are counted. Formula (BF.38) already treats the two-point reflection orbit.

In particular, a reflection-stable actual packet containing an off-line root forces a strictly positive right side of (BF.36) at every finite level. Then W_n is nonzero, indefinite, and of rank exactly two, with generalized eigenvalues $\epsilon_n, -\epsilon_n$. The strict positivity follows from the packet sum; both signs and exact rank follow from (BF.25) and (BF.16). For an arbitrary packet without reflection stability, the valid statements are instead the separate inequalities (BF.35); its trace need not vanish.

This lower bound is on the relative expression containing G_n^{-1} . The absolute monotonicity (BF.17) does not supply an upper bound on that inverse-weighted expression. The homotopy freedom in A1771, lines 4432–4475, does not alter this conclusion. Retain the Fourier transform

$$(\mathcal{F}\phi)(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

It restricts to an isomorphism $V \rightarrow V$: it preserves evenness and the Schwartz space, and interchanges evaluation at zero with the integral, so preserves the two zero moments in (BF.01). Its inverse on V is the inverse Fourier transform with the same convention. For the two-leg differential and its cycle map,

$$\begin{aligned} d_{\text{leg}} : V \oplus V &\longrightarrow \mathcal{B}, & d_{\text{leg}}(v, w) &= \Theta(v - \mathcal{F}w), \\ \gamma : V &\longrightarrow V \oplus V, & \gamma(v) &= (v, \mathcal{F}^{-1}v), \end{aligned}$$

one has $d_{\text{leg}}\gamma = 0$ by direct substitution. For a linear map $L : E_Z \rightarrow V$, adding γL to a primitive therefore leaves B_n unchanged, and hence leaves W_n unchanged for fixed R_n . The support transition described above likewise records where the boundary becomes a supported zero while retaining the pairing that produced (BF.16).

There is also an exact tensor obstruction independent of the rank-two argument. For an integer $k \geq 1$, keep

$$A_k = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(k-j)} \quad \text{on } E_Z^{\otimes k}. \quad (\text{BF.40})$$

For any positive definite metric H on this same tensor space, set $W_{k,H} = A_k^* H + H A_k - kH$. The nonzero tensor $v^{\otimes k}$ of the eigenvector above satisfies $A_k v^{\otimes k} = k\rho v^{\otimes k}$. Consequently

$$\frac{(v^{\otimes k})^* W_{k,H} v^{\otimes k}}{(v^{\otimes k})^* H v^{\otimes k}} = k(2\operatorname{Re} \rho - 1). \quad (\text{BF.41})$$

The denominator is strictly positive. Thus the least two-sided excess for $H, W_{k,H}$ is at least $k|2\operatorname{Re}\rho - 1|$. In particular this applies to the actual orbit metric of A1721, lines 4102–4139,

$$H_{k,T} = \int_0^T e^{-kt} \exp(tA_k)^* G_n^{\otimes k} \exp(tA_k) dt, \quad T > 0.$$

It is positive definite because the integrand has positive quadratic form on every nonzero vector at every t ; $\exp(tA_k)$ is invertible. The same time parameter is used in every factor, and the full weight e^{-kt} remains. Changing T cannot change the Rayleigh value (BF.41).

For the empty packet, $h_Z = 1$, $E_Z = \{0\}$, and every packet sum is zero. Define its control excess to be zero. Then the boundary-floor assertion is the equality $0 = 0$. The extension row $[t^{d-1}]R_Z$ and the positive-dimensional Gram inverses above are not invoked in this empty case.

Equations (BF.35) and (BF.36) are lower bounds proved on the actual arithmetic objects. They neither assert an off-line zero nor provide the arithmetic upper estimate needed to exclude one. The source explicitly retains that missing estimate in A1721, line 4259, and A1771, line 4890. Finite regression checks, the stated integer-sum tail bound, and absolute Gram decrements do not supply it.